claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-a91026eafb9f34264.jsonl`
- SHA-256: `e9e1962d8a4f64cbfabd8996d10f3e321a09411f220f230142c801e2b6173a6f`
- Source modified: `2026-06-06T17:31:28+00:00`
- Imported at: `2026-07-05T16:48:28+00:00`
- project: `wf_6237b110-d58`
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`

Transcript

1. user (2026-06-06T17:30:26.463Z)

Source Extractor

```
Research question: "Research alternative methods to automatically detect rally start/end
segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-
HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation
is the best approach or whether other methods are competitive or complementary. Keep it focused
("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches
over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):
- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY
inference at runtime ? privacy lever: user video never leaves the device. It's a commercial
product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-
commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We
derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We
ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision
ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label
rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features).
Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the
learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck
(its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes /
pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd
noise, umpire/score calls).

SPECIFICALLY COVER:
1. Method families for rally/point segmentation in racket sports (badminton, tennis, table
tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep
models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance"
gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-
overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM
teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference
COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their
licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher
? local-distillation the best, or are some alternatives competitive or complementary ?
especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run
fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory +
person detector + audio)."
```

Fetch and extract key claims from this source:
URL: https://arxiv.org/pdf/2307.16000
Title: Automated Hit-frame Detection for Badminton Match Analysis (arXiv 2307.16000)
Found via: classical/lightweight search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
   - be a concrete, checkable statement (not vague generalities)
   - include a direct quote from the source as support
   - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-06T17:30:28.813Z)

I'll fetch the source and extract the relevant claims.

3. user (2026-06-06T17:30:39.617Z)

I appreciate the request, but I need to be transparent: the content provided appears to be a **corrupted or improperly encoded PDF file**. The text shown is a stream of compressed/encoded data that cannot be meaningfully parsed.

What I can observe:
- The data begins with PDF markers (`%PDF-1.5`)
- The content contains FlateDecode streams (binary compressed data)
- No readable text about badminton, hit-frame detection, shuttle tracking, or related methodology is extractable

To properly assist, I would need:
1. A clean, readable PDF or text version of the paper
2. The actual paper title and authors
3. A version that hasn't been corrupted in transmission

If you have access to the original paper on automated hit-frame detection for badminton, please share:
- The title
- A readable version (ASCII text or properly formatted PDF)
- Or a link to the published version

I'm ready to extract the specific information you requested (methods, accuracy metrics, datasets, quotes, publication details) once readable source material is available.

[Binary content (application/pdf, 1.5MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\tool-results\webfetch-1780766977975-dnjzkl.pdf]

4. assistant (2026-06-06T17:30:42.344Z)

The PDF was downloaded but not parsed by WebFetch. Let me read it directly with the Read tool, which can handle PDFs.

5. user (2026-06-06T17:30:43.289Z)

pdftoppm failed: Command 'pdftoppm' not found or is in an unsafe location (current directory)

6. assistant (2026-06-06T17:30:46.571Z)

The Read tool needs pdftoppm for PDF rendering. Let me try the abstract page on arxiv instead, which is HTML.

7. user (2026-06-06T17:30:55.631Z)

Automated Hit-frame Detection for Badminton Analysis - Information Extraction

Based on the abstract page provided, here's what can be extracted:

1. Hit-frame Detection Method

The abstract indicates a multi-step approach but doesn't specify exact technical details. It mentions "modern deep learning techniques" and includes:
- Rally-wise video trimming
- Player and court keypoints detection
- Shuttlecock flying direction prediction
- Hit-frame detection

2. Quantitative Results Reported

- 99% accuracy on shot angle recognition for video trimming
- Over 92% accuracy for player keypoints sequences applied to shuttlecock flying direction prediction
- "evaluation results of rally-wise video trimming and hit-frame detection" mentioned but not specified in the abstract

3. Datasets and Licenses

Not mentioned in the abstract page provided.

4. Shuttle Trajectory Detector

Not specified in the abstract page.

5. Publication Details

- **Paper ID:** arXiv:2307.16000
- **Authors:** Yu-Hang Chien, Fang Yu
- **Submission:** July 29, 2023 (v1); August 2, 2023 (v2)
- **Category:** Computer Vision and Pattern Recognition (cs.CV)

6. Direct Quotes

"The proposed approach in this study comprises several automated procedures like rally-wise video trimming, player and court keypoints detection, shuttlecock flying direction prediction, and hit-frame detection."

"We achieved 99% accuracy on shot angle recognition for video trimming, over 92% accuracy for applying player keypoints sequences on shuttlecock flying direction prediction."

8. assistant (2026-06-06T17:30:58.413Z)

I have solid extraction from the abstract page. Let me get a bit more detail on the method and conclusions by fetching the HTML version if available.

9. user (2026-06-06T17:31:06.014Z)

Automated Hit-frame Detection for Badminton Match Analysis

Full Abstract

"Sports professionals constantly under pressure to perform at the highest level can benefit from sports analysis, which allows coaches and players to reduce manual efforts and systematically evaluate their performance using automated tools. This research aims to advance sports analysis in badminton, systematically detecting hit-frames automatically from match videos using modern deep learning techniques. The data included in hit-frames can subsequently be utilized to synthesize players' strokes and on-court movement, as well as for other downstream applications such as analyzing training tasks and competition strategy. The proposed approach in this study comprises several automated procedures like rally-wise video trimming, player and court keypoints detection, shuttlecock flying direction prediction, and hit-frame detection. In the study, we achieved 99% accuracy on shot angle recognition for video trimming, over 92% accuracy for applying player keypoints sequences on shuttlecock flying direction prediction, and reported the evaluation results of rally-wise video trimming and hit-frame detection."

Key Findings

Inputs for hit-frame detection: Player and court keypoints, along with shuttlecock trajectory data

Shuttle direction prediction: Yes?the abstract explicitly states the method achieved "over 92% accuracy for applying player keypoints sequences on shuttlecock flying direction prediction," indicating player pose drives shuttle direction estimation

Overall goals: The research targets automated badminton match analysis to enable coaches and players to "reduce manual efforts" while evaluating performance systematically, with applications in strategy and training task analysis

10. assistant (2026-06-06T17:31:09.933Z)

I have enough to extract falsifiable claims. This is a primary research source (arXiv preprint by Yu-Hang Chien and Fang Yu, cs.CV). Now I'll synthesize the claims relevant to the research question on rally segmentation for a self-hosted, local-first badminton pipeline.

11. user (2026-06-06T17:31:19.335Z)

Structured output provided successfully

12. assistant (2026-06-06T17:31:28.016Z)

I've completed the source extraction. The structured output has been submitted successfully.

Summary of findings from arXiv:2307.16000 "Automated Hit-frame Detection for Badminton Match Analysis" (Chien & Yu, 2023):

- **Source quality**: primary (arXiv cs.CV preprint, badminton-specific)
- **Key relevance to the research question**: This is a strong data point for the **pose/player-stance + court-homography state machine** family. It localizes shots (hit-frames) and trims rallies using player keypoints and court keypoints, and notably predicts shuttlecock flying direction from **player pose sequences (>92% accuracy)** rather than relying solely on a shuttle tracker ? directly relevant to your team's open question of whether pose/court-geometry can complement or de-risk the WASB shuttle-trajectory bottleneck.
- **Caveats for falsifiability**: The headline numbers (99% shot-angle recognition; >92% pose-to-direction) are component-level accuracies, not end-to-end rally-segmentation F1 at a given IoU, so they aren't directly comparable to your F1?0.44 ceiling. The abstract did not disclose datasets/licenses or the specific shuttle detector, and the full-text PDF could not be parsed in this environment (no pdftoppm; WebFetch returned raw PDF bytes), so dataset/license claims could not be verified from primary text.